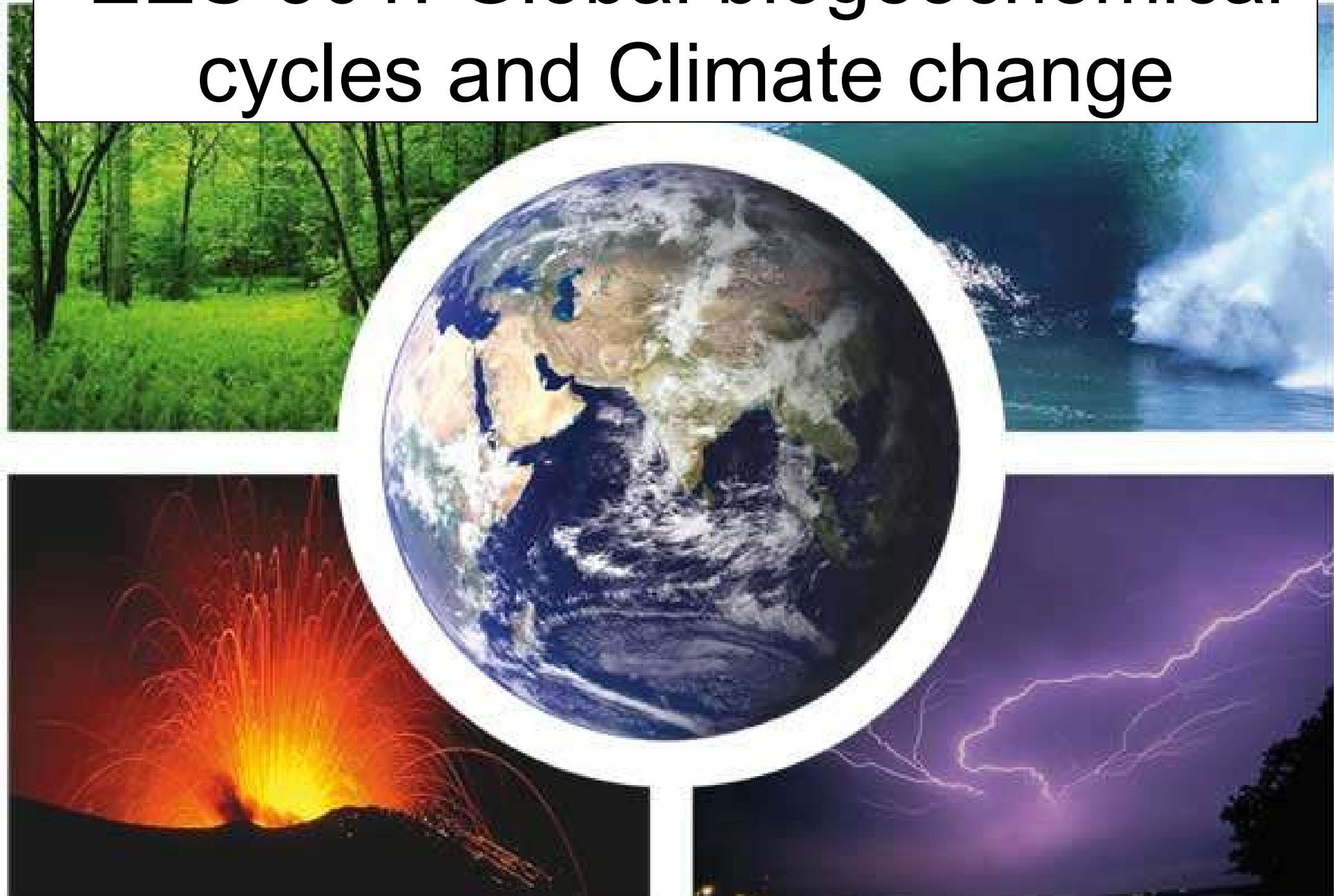


EES 301: Global biogeochemical cycles and Climate change



Conditions far removed from standard temperature and pressure conditions

- While dealing with reactions in the stratosphere at < 200 K and less than 0.1 bar or with reactions between hydrothermal fluids circulating at mid-ocean ridges and rocks, we need values for ΔH and ΔG that are very far from the standard temperature and pressure conditions for which numbers can be found in tables.
- For many simple molecules the functions $(G-H)/T$ and $(H-H_{298})/T$ are tabulated for extensive temperature ranges. The value for the desired temperature can be interpolated from such tables.
- We can also use the basic thermodynamic relationships between G and H and T and P to derive values. For example the heat capacity at constant pressure (C_p) of a substance is defined as.

$$C_P = \left(\frac{\delta H}{\delta T} \right)_P$$

- From this we can obtain an expression for the temperature dependence of ΔH of a reaction if the heat capacity at constant pressure is known

Conditions far removed from standard temperature and pressure conditions

•For the pressure dependence the following fundamental relationship can be used:

$$\left(\frac{\delta H}{\delta P}\right)_T = V(1 - \alpha T)$$

•In which α is the coefficient of thermal expansion:

$$\alpha = \left(\frac{1}{V}\right) \left(\frac{\delta V}{\delta T}\right)_P$$

Non-ideal Behavior

- Most gases and gas mixtures in the atmosphere can be approximated as ideal gases.
- In the environment departures from ideality are common in the condensed phase.
- A significant departure from ideality results from the effect of the ionic strength of aqueous solutions on the energies of ions. This needs to be considered while looking at equilibria in ocean water and brines.

Thermodynamic isotope effects

- Thermodynamic energy terms (and equilibrium constants) may differ for compounds containing different isotopic species of an element.
- An important example is the case of the vapor/liquid equilibrium for water. The vapor pressure of a lighter isotopic species like H_2^{16}O (18 amu) is higher than that of heavier species like HD^{16}O , H_2^{17}O (19 amu) and H_2^{18}O (20 amu).
- The differences are very small hence small variations in the isotopic composition are described by comparing the ratio of isotopes in the sample material to the ratio of isotopes in a reference material in parts per thousand (per mil denoted by ‰)

$$\delta^{18}\text{O} = \left[\frac{\left(\frac{^{18}\text{O}}{^{16}\text{O}} \right)_{\text{sample}}}{\left(\frac{^{18}\text{O}}{^{16}\text{O}} \right)_{\text{ref}}} - 1 \right] \times 1000$$

For ^{18}O and D the standard is Vienna Standard Mean Ocean water distributed by the IAEA

Thermodynamic isotope effects

- If a sample has less of the reference isotope (^{18}O) then the delta is negative and the sample is said to be light.
- The ratio of the equilibrium vapor pressure of ^{16}O water P_{16} and ^{18}O water P_{18} depends on temperature and is expressed by the following equation

$$\ln\left(\frac{P_{16}}{P_{18}}\right) = \frac{7.88}{T} - 0.0177$$

- At $25^\circ\text{C} = 298\text{ K}$ the ratio is $1.0088 = e^{\left(\frac{7.88}{298} - 0.0177\right)}$

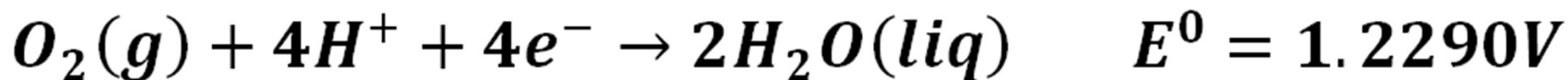
- This means that the $^{18}\text{O}/^{16}\text{O}$ ratio of the atmosphere is that of the ocean multiplied with $1/1.0088$. Expressed in permil the atmosphere has a $\delta^{18}\text{O}$ of -8.7‰ .
- This means water vapor in the atmosphere is “lighter” than the ocean from which it evaporated and when the water moves upwards via convection and the vapor condenses, the rain formed is heavier than the vapor from which it formed. This makes the vapor that is transported land inwards by winds even lighter. But rain formed will always be heavier than the gas phase still present by a factor that is determined by the temperature at which it forms

Oxidation & Reduction: Nernst Equation

- Reactions in which electrons are transferred from one reactant to another are called oxidation-reduction or redox reactions. In such reactions we focus on the transfer of electrons when examining conditions for spontaneity and equilibrium because electrons are the common currency through which different reactions can be compared.

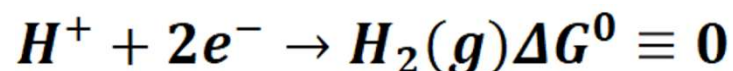
- The elemental reaction used to describe a redox reaction is the half reaction usually written as a reduction.

- In the reduction of oxygen atoms O_2 (oxidation state 0) to H_2O (oxidation state -2) the half-cell potential E^0 is given in volts after the reaction:



Oxidation & Reduction: Nernst Equation

- The half-cell free energy change ΔG^0 is a quantity that gives the free energy change for the given reaction at standard conditions relative to the reduction for the standard hydrogen electrode for which the half reaction is:



- The half-cell potential and the half-cell free energy change ΔG^0 are related by the following relationship for reversible conditions

$$\Delta G^0 = -zFE^0$$

F= Faraday constant (=94490 C/mol or 94490 J/V/mol and z is the number of electrons appearing in the balanced reduction half cell.

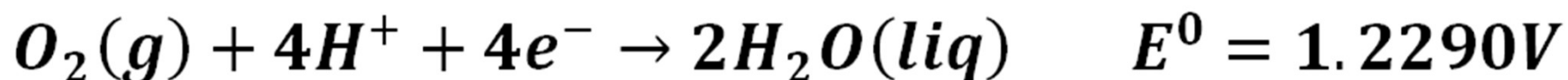
- A negative value for ΔG^0 or a positive value for ΔE^0 means the reduction has a greater tendency to occur than the reduction of H^+ to H_2

Oxidation & Reduction: Nernst Equation

- The Nernst equation describes the dependence of the half-cell potential on concentration:

$$E = E^0 - \left(\frac{2.303RT}{zF} \right) \log \left(\frac{\text{Products}}{\text{Reactants}} \right)$$

- The symbol E_h is often used to make clear that we are representing a half-cell potential relative to the standard hydrogen electrode.
- For the reduction of oxygen for which the E^0 is 1.2290 V the Nernst equation under standard temperature pressure conditions for which the factor $2.303RT/F=0.03916$ comes out to be



$$E_h = 1.2290 - \left(\frac{0.05916}{4} \right) \log \left(\frac{1}{P_{O_2}[H^+]^4} \right)$$

Oxidation & Reduction: Electron activity

- As an alternative the tendency for a reaction to occur may also be expressed in terms of a hypothetical electron activity based on the standard hydrogen electrode. The free energy of an electron is related to chemical activity of the electron α_e by:

$$G = G^0 + RT \ln(\alpha_e)$$

- The free energy change for a process in which z electrons move from a standard hydrogen electrode α_{SHE} to some other electrode is therefore:

$$\begin{aligned}\Delta G^0 &= [\Delta G^0 + RT \ln(\alpha_e)] - [\Delta G^0 + RT \ln(\alpha_{SHE})] \\ &= zRT \ln(\alpha_e / \alpha_{SHE})\end{aligned}$$

- The z comes from the fact that the potential required to move $1e$ α_e is related to the materials involved in the reaction. How many e you move in the actual reaction is accounted for in z . Same as in: $\Delta G^0 = -zFE^0$

- We define the term P_ϵ related to electron activity as:

$$P_\epsilon = -\log(\alpha_e) = -\Delta G / (2.303zRT) = E_h(F/2.303RT)$$

The 2.303 comes from changing the $\ln$ to a $\log$. Electrons move from a more negative to a more positive P_ϵ spontaneously. It is related to concentrations as follows:

$$P_\epsilon = P_\epsilon^0 - \left(\frac{1}{z}\right) \log \left(\frac{[products]}{[reactant]} \right)$$

P_ε /pH stability diagram

•The reaction $O_2(g) + 4H^+ + 4e^- \rightarrow 2H_2O(liq)$ $E^0 = 1.2290V$

can be rewritten as: $P_\varepsilon = P_\varepsilon^0 - \left(\frac{1}{4}\right) \log \left(\frac{1}{P_{O_2}[H^+]^4} \right)$

•If we consider an oxygen pressure of 1 bar then considering the definition of $pH = -\log [H^+]$ and using the relationship between P_ε^0 and E^0 (half cell potential in volts)

$$P_\varepsilon = 20.77 - pH$$

•While for the reduction of H^+ to H_2 at an hydrogen pressure of 1 bar

$$P_\varepsilon = -pH$$

P_ϵ /Ph stability diagram

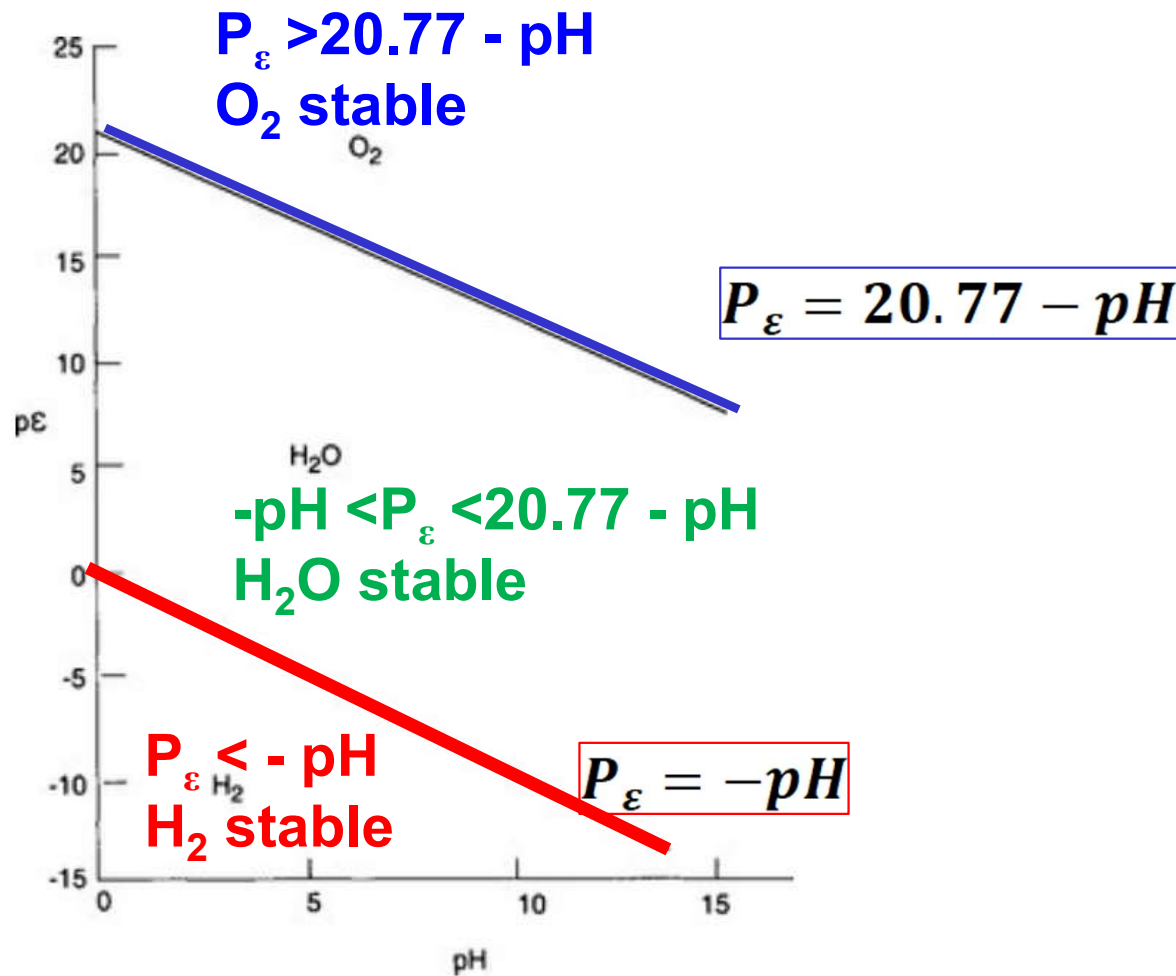


FIG. 5-1 Stability diagram for H₂O.

Kinetics

While Thermodynamics informs us about the feasibility of a process or reaction, it tells us nothing about how fast the process or reaction will occur!

Kinetics is the study of how fast or slow a process or reaction occurs. For determining the fate of chemicals in the environment through perturbed biogeochemical cycles (e.g. lifetime of green house gases, degradation of water pollutants) as well as in natural biogeochemical cycles, it is Kinetics that is even more important than thermodynamics. After all, our atmosphere is full of N_2 and O_2 but has very little NO because the reaction $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$ has a high activation energy and is very slow under ambient conditions.

In chemical kinetics an empirical relationship is used to relate the overall rate of a process to the concentrations of various reactants:

$$dA/dt = k[A]^n[B]^m$$

m and n represent the order of the reaction with respect to A and B and the overall reaction order is the sum of these exponents = $n + m$

Kinetics

For a simple reaction $A \rightarrow B$ $rate = k[A]^n$

- n is the reaction order – usually 0, 1, or 2 and determined by how the reaction rate depends on the concentration of A
- k is the rate constant. This constant can only be measured in the lab

$n = 0$	$dA/dt = k[A]^0 = k$	zero order
$n = 1$	$dA/dt = k[A]^1 = k[A]$	first order
$n = 2$	$dA/dt = k[A]^2$	second order

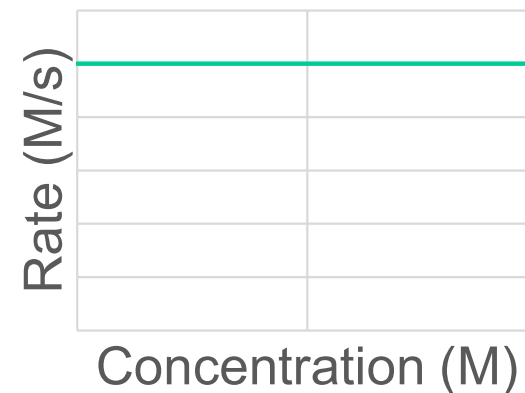
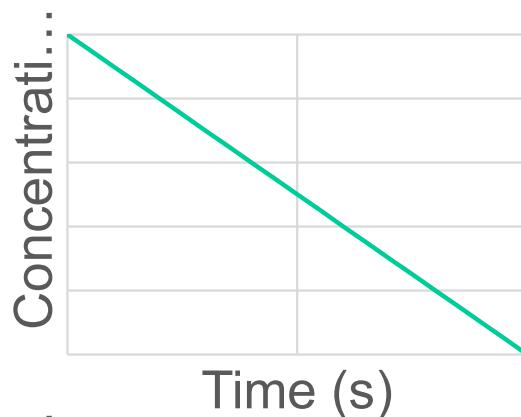
An example of a simple first order reaction would be radioactive decay like: $^{14}\text{C} \rightarrow ^{14}\text{N} + e^-$ in such a case $d[^{14}\text{C}]/dt = -k[^{14}\text{C}]$ with a k constant in the units $1/s$ which we use in the form $\ln[^{14}\text{C}]_t = -kt + \ln[^{14}\text{C}]_0$ for dating

Photolytic reactions such as the decomposition of O_3 by light or also unimolecular processes $\text{O}_3 + h\nu \rightarrow \text{O}_2 + \text{O}$

Two graphs – *conc. vs. time* and *rate vs. conc.*

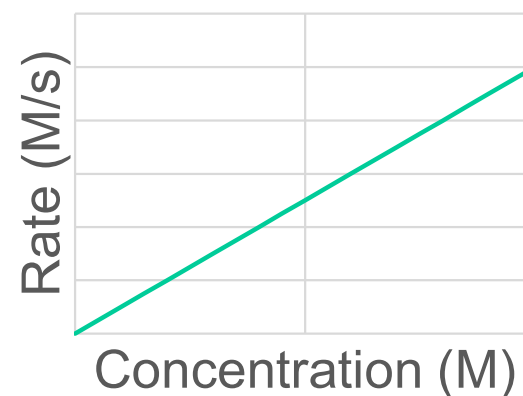
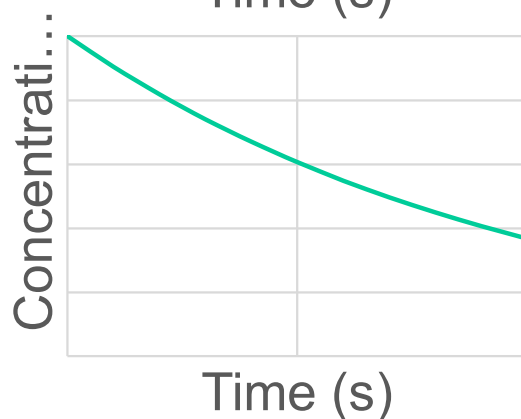
Zero order

- $rate = k$
- If $[A]$ changes, rate remains constant



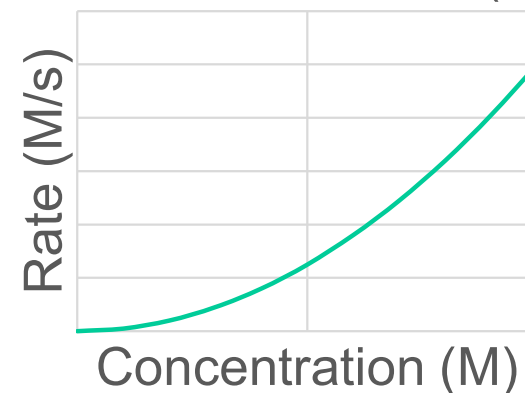
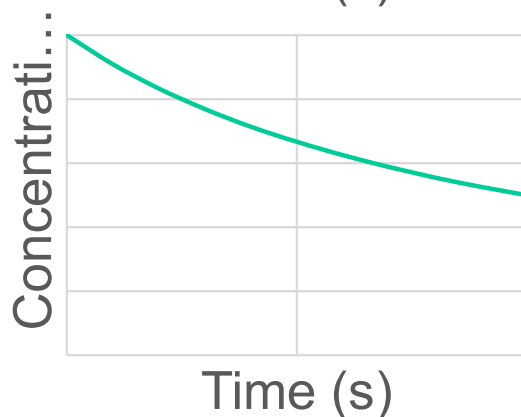
First order

- $rate = k[A]$
- If $[A]$ doubles, rate doubles



Second order

- $rate = k[A]^2$
- If $[A]$ doubles, rate quadruples



1st order reaction rates $[A]$ vs t is an exponential but $\ln[A]$ vs time becomes a line

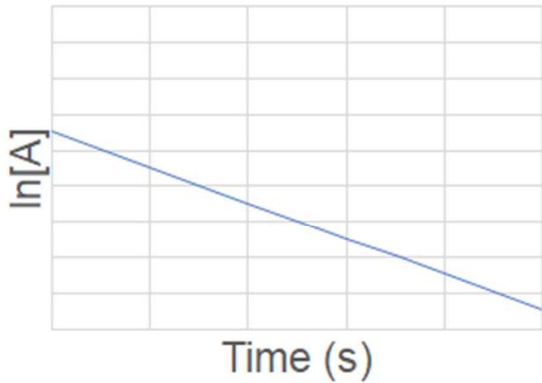
$$\ln[A]_t = -kt + \ln[A]_0$$

↑ ↑ ↓

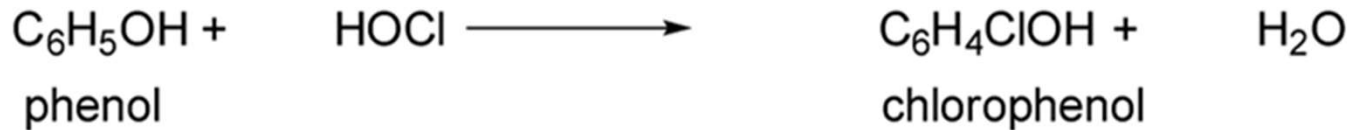
variable constant variable constant

$$y = mx + b$$

↑ ↑ ↑ ↑
variable constant variable constant



Slope = $-k$
Intercept = $\ln[A]_0$



The chlorination of phenol is a pseudo first-order reaction (i.e., first order under the right conditions) studied in water-treatment systems

Radioactive decay as 1st order reaction: the half life concept

The half life of a substance is the amount of time it takes concentration to be reduced to half of its original value. For a first-order reaction:

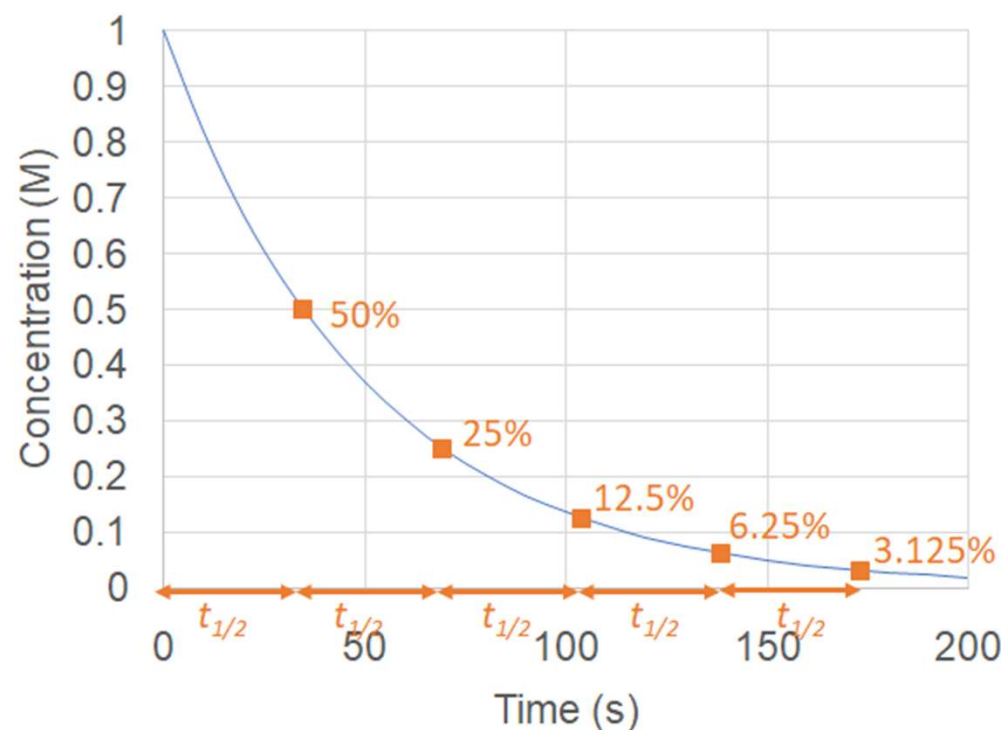
$$\ln \frac{[A]_t}{[A]_0} = -kt$$

$$\text{If } [A]_0 = 100 \text{ and } [A]_t = 50: \ln \frac{50}{100} = -kt$$

$$-0.693 = -kt$$

$$t_{1/2} = \frac{0.693}{k}$$

In other words, the half life of a first-order reaction depends only on k and does not depend on the starting concentration.



Photolysis as 1st order reaction:

$$X + h\nu \rightarrow \dots \quad k = \int_0^\infty \underbrace{q_X(\lambda)}_{\text{quantum yield}} \underbrace{\sigma_X(\lambda)}_{\text{absorption X-section}} \underbrace{I_\lambda}_{\text{photon flux}} d\lambda$$

Quantum yield: Probability that after absorption of photon, product will be formed, values = 1 mean

Every reactant molecule that absorbs the relevant photons will photo-dissociate to form products

Absorption cross-section: Probability that the molecule will absorb photons of given wavelength [cm²/molecule]

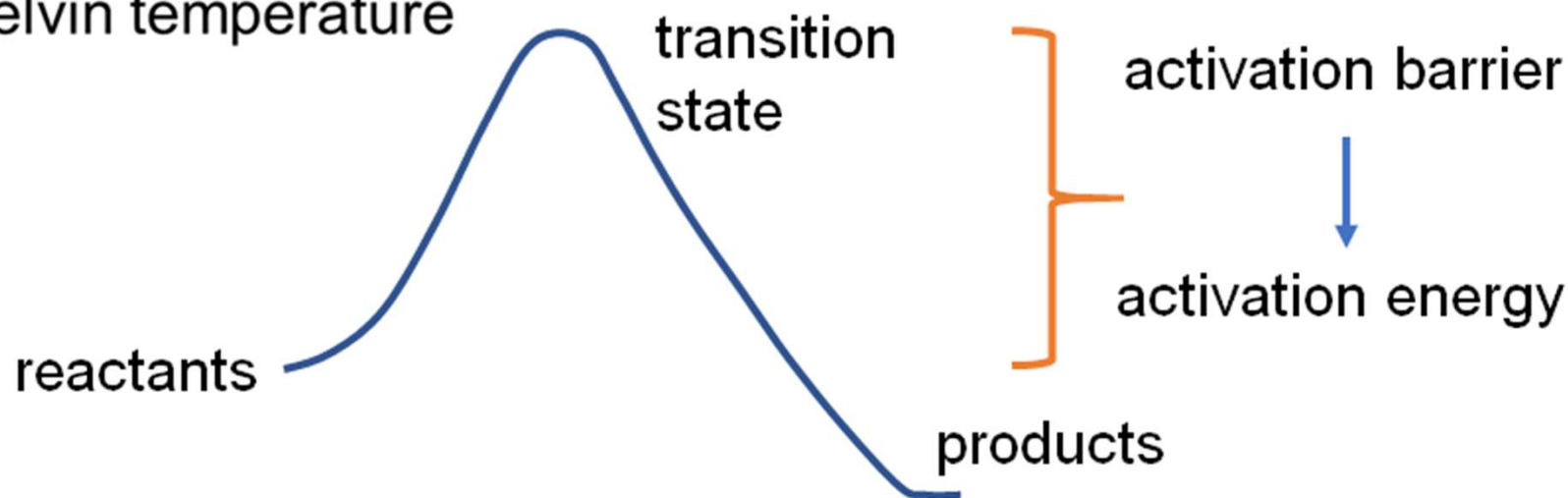
e.g. ozone has strong absorption in UV region of EM spectrum

Photo flux: Amount of incident photon flux [photons cm⁻² s⁻¹]

Temperature dependence = Arrhenius equation

$$k = Ae^{-E_a/RT} \quad \text{- Arrhenius equation}$$

- A – frequency factor or pre-exponential factor
- E_a – activation energy
- R – gas constant ($8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)
- T – Kelvin temperature



$\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$ takes less than 1 s at more than 2000 K. This is why NO gets produced in motors, power plants and lightening flashes

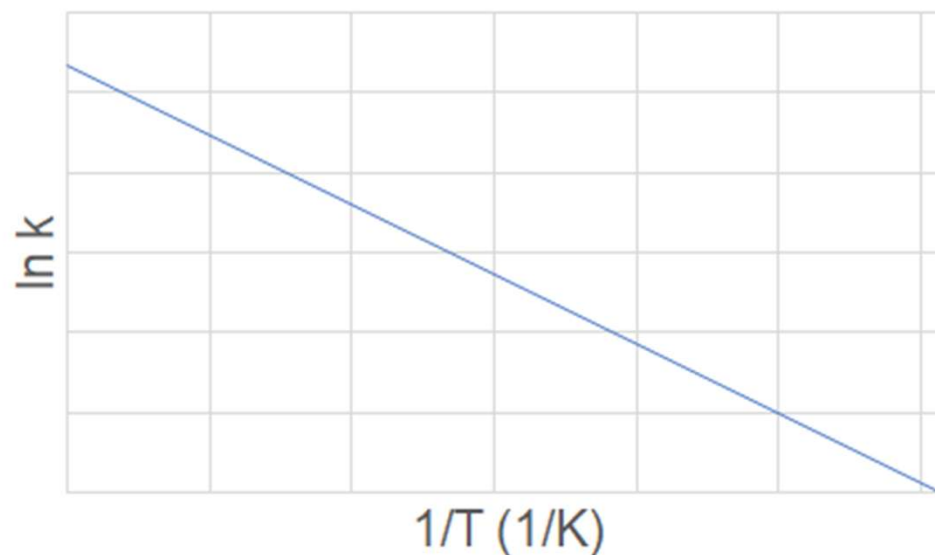
Temperature dependence = Arrhenius equation

$$k = Ae^{-E_a/RT}$$

$$\ln k = \ln(Ae^{-E_a/RT})$$

$$\ln k = \ln A + \ln e^{-E_a/RT}$$

$$\ln k = \ln A - E_a/RT$$



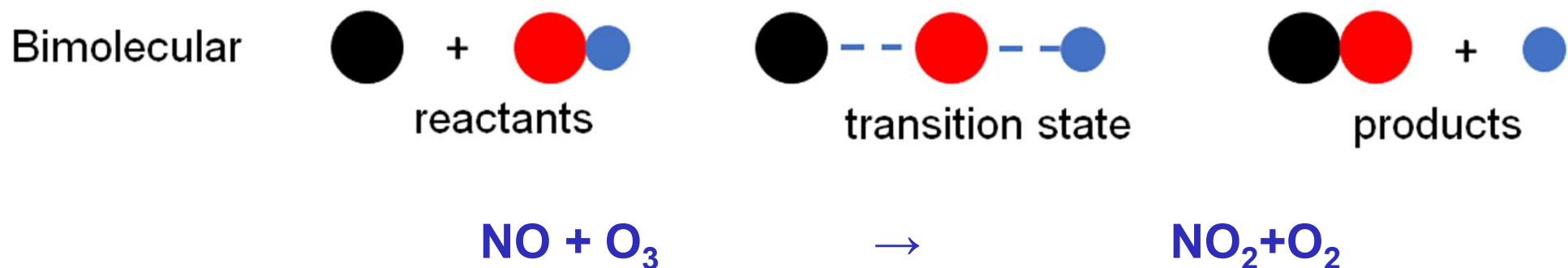
$$\underbrace{\ln k}_{\text{variable}} = -\frac{E_a}{R} \left(\underbrace{\frac{1}{T}}_{\text{variable}} \right) + \underbrace{\ln A}_{\text{constant}}$$

variable constant variable constant

$$\text{Slope} = E_a/R$$

$$\text{Intercept} = \ln A$$

Bimolecular reactions



In bimolecular reactions the rate depends on the product of concentrations of the two reactants

$$-\frac{d[\text{NO}]}{dt} = -\frac{d[\text{NO}_2]}{dt} = -k[\text{NO}][\text{O}_3]$$

The rate constant k_a for a bimolecular reaction has the units $[\text{cm}^3/\text{molecule s}]$

termolecular reactions

Termolecular processes are common when two small reactant molecules combine to form a single small molecule



These reactions are often very exothermic and the role of the 3rd partner, which in the atmosphere is often N₂ is to carry away some of the excess energy to ensure the product doesn't break apart immediately

$$d[\text{O}_3]/dt = -k[\text{O}][\text{O}_2][\text{M}]$$

Since three bodies need to collide, these reactions are extremely pressure dependent. The units for the termolecular rate constant are [cm⁶/molecule s]

PRACTICE QUESTIONS

**You will have to ask your doubts in this weeks tutorial Friday
29th August**

Long answer & numericals

1. What is the molarity of a 1 L solution that contains 156 g of nitric acid HNO_3
2. Sea water contains 2.7 g NaCl per 100 mL
 - a) What is the percentage concentration of NaCl expressed in weight %
 - b) What is the molarity of NaCl in seawater
3. The molarity of MgCl_2 in seawater is 0.54
 - a) How many grams of MgCl_2 are present in every 100 mL of sea water
 - b) What is the concentration of MgCl_2 expressed in weight %
 - c) How many mmol/mL of MgCl_2 does seawater contain?
4. Your guide asks you to prepare 1 L of 0.1 molar KNO_3 solution. How many g of KNO_3 do you need to weigh into 1 L of water to accomplish that?
5. If 100 g water contains 1.5 mg of arsenic what is the concentration of arenic in ppm.

Long answer style & numericals

1. The ratio of ^{18}O to ^{16}O in the ocean water is about 2045:100000.
 - a) Suppose 10^6 atoms of ^{16}O evaporate in equilibrium with ocean water at 298 K. How many ^{18}O atoms will be associated with these ^{16}O atoms in the atmosphere?
 - b) Suppose now that this air is shifted to higher latitudes and cooled to 293 K wherein half of the ^{16}O atoms condense to form raindrops. What number of ^{18}O atoms will condense and what number will remain in the atmosphere
 - c) Evaluate the $\delta^{18}\text{O}$ for the precipitation that forms in (b) as well as the $\delta^{18}\text{O}$ of the gas phase water left behind in the air parcel that moves further north.
 - d) Describe verbally what will happen to the rain that forms inside a continent as the air mass moves away from the ocean and the moisture rains out bit by bit while the air mass moves inland

Long answer style & numericals

1. A first-order reaction is found to have a rate constant, $k=5.5 \times 10^{-5} \text{ s}^{-1}$. What is the half life of the substance in question.
2. A first-order reaction has a rate constant, $k=4 \times 10^{-3} \text{ s}^{-1}$ and the concentration of the reactant is 0.02 molar. Calculate the rate of the reaction.
3. If a zero order reaction has a reactant of 0.1 molar concentration and a rate coefficient of , $k=1 \times 10^{-2} \text{ s}^{-1}$, how long will the reaction take to go to completion?
4. State two quantities that must be measured to establish the rate of a chemical reaction and cite several factors that affect the rate of a chemical reaction. How do you plot them to obtain k ?
5. State the quantities that must be measured to establish the Arrhenius equation for a reaction. How do you plot these to obtain the parameters of the equation and how would you calculate the activation energy from such a plot?
6. For the reaction, $A \rightarrow \text{products}$, a graph of $[A]$ versus time is a curve while rate vs concentration is a straight line . What can be concluded about the order of this reaction?

Long answer style & numericals

1. In the reaction $A \rightarrow \text{products}$, we find that when $[A]$ has fallen to half of its initial value, the reaction proceeds at the same rate as its initial rate. Is the reaction zero order, first order, or second order? Explain.
2. Following are two statements pertaining to the reaction $2A + B \rightarrow 2C$, for which the rate law is $\text{rate} = k[A][B]$. Identify which statement is true and which is false, and explain your reasoning.
 - (a) The value of k is *independent* of the initial concentrations $[A]_0$ and $[B]_0$.
 - (b) The unit of the rate constant for this reaction can be expressed either as s^{-1} or min^{-1} .
3. In the reaction, $A \rightarrow \text{products}$, with the initial concentration $[A]_0 = 1.512 \text{ M}$, $[A]$ is found to be 1.496 M at $t = 30 \text{ s}$. With the initial concentration $[A]_0 = 2.584 \text{ M}$, $[A]$ is found to be 2.552 M at $t = 1 \text{ min}$. What is the order of this reaction?